Attach this Cover Page as the first page of your report PDF submission. Answer in your own words; keep each answer to the space provided.

A. Student and team particulars

Name / Student ID	Wang Youyang / G2606821F
Undergrad University / Degree	UESTC / Electronic Information Engineering (Bachelor)
Partner's name and ID (if any)	None (individual project)
Tentative Plan after NTU Master	Pursue a PhD

B. Links (leave blank if not applicable)

Your CV Online and Github page	https://github.com/youyang0313 (profile page with CV)
Online project repository (e.g., Github code, data, tools)	https://github.com/youyang0313/ecg-lead-masking
LecTrace Script/Demo Website (eg. https://praisegee.github.io/lectrace-demo)	https://youyang0313.github.io/ecg-lead-masking/
Overleaf / report link (optional)	
Demo link, eg. video (optional)	

Most relevant software tool in project	PyTorch
Your project extends Assignments 1 or 2?	No

Describe the software bug in your deep learning program (e.g., Pytorch, Tensorflow or ConvNetJS Javascript etc) that cost you the most time. How did you find it, and how did you know it was fixed? Did AI coding agent help you to find bugs?
In the lead-subset extension, missing limb leads were reconstructed only when leads I and II were both present, although any two limb leads suffice. The AI-simulated peer review found it. I confirmed the fix by a 0.3-0.5% relative reconstruction error and the expected change in downstream metrics; AI helped write the verification script.

Which of the Lecture Videos (https://www.youtube.com/playlist?list=PLORG7zBQosxw) can be improved and why?
I would suggest improving Lecture 8 by adding a short, end-to-end healthcare example to the contrastive-learning section (slides 49–84): a worked example showing how to construct positive and negative pairs, calculate the loss, and reuse the encoder for a downstream task would make the connection between theory and implementation clearer.

C. LecTrace — execution-explanation presentation

Approximate time spent learning LecTrace	About 3 hours
LecTrace script written by yourself or mostly by AI?	Mostly by AI (Claude Code), directed and reviewed by me

Q. Which execution states did you choose to show, and why those rather than others?
(1) A zero-filled input breaking Einthoven's law, (2) BatchNorm input statistics drifting inside the forward pass, (3) re-estimating BatchNorm to repair the model without retraining, (4) the result across 207 lead subsets. Together they show why the model fails and when masking still helps; plain accuracy tables would not.

Q. What was confusing or missing in LecTrace, and what one feature would you improve or add?
Functions called from main() serialize every local variable, so a large test array made one build take over 10 minutes. I would add a size limit, or inspect only marked variables by default.

By default, you consent to your anonymised answers being used in research on teaching. Let me know if otherwise. This choice doesn't affect your grade.